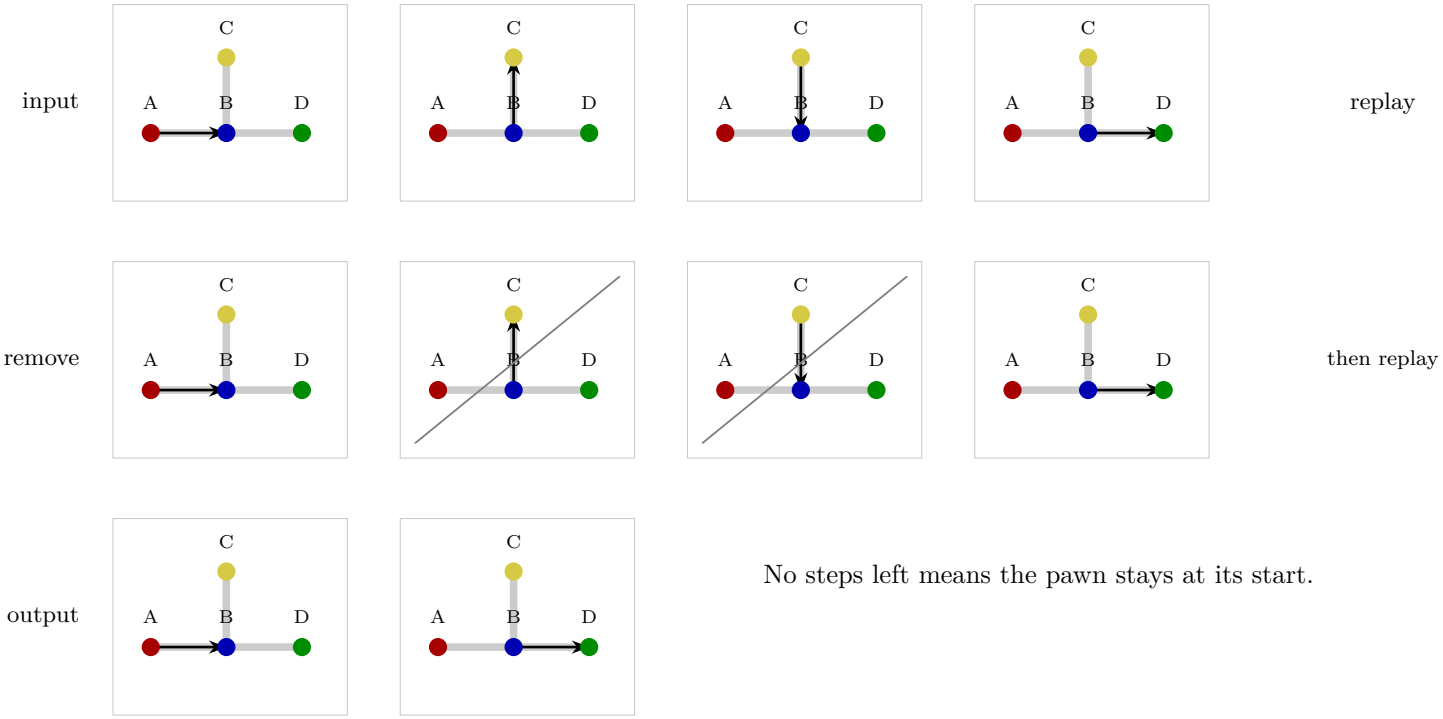
Move a pawn on the roads. Keep each journey as ordered step tiles; an adult may record. Remove only two adjacent steps along the same road in opposite directions. Replay the other steps in order. The map, start, and finish stay fixed.



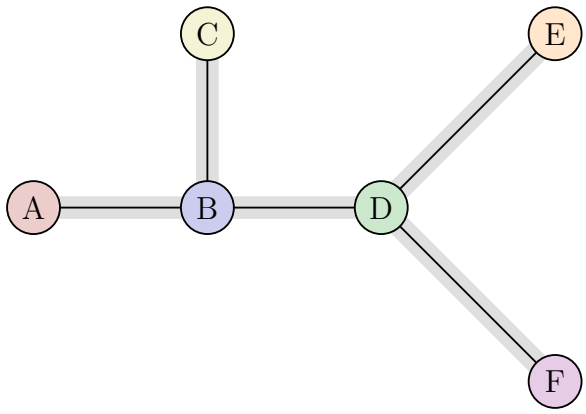
No steps left means the pawn stays at its start.

Problem 1: Shorten each recorded trip until no move is possible. Make a different seven-step trip from C to F that finishes with the same route as the last one.

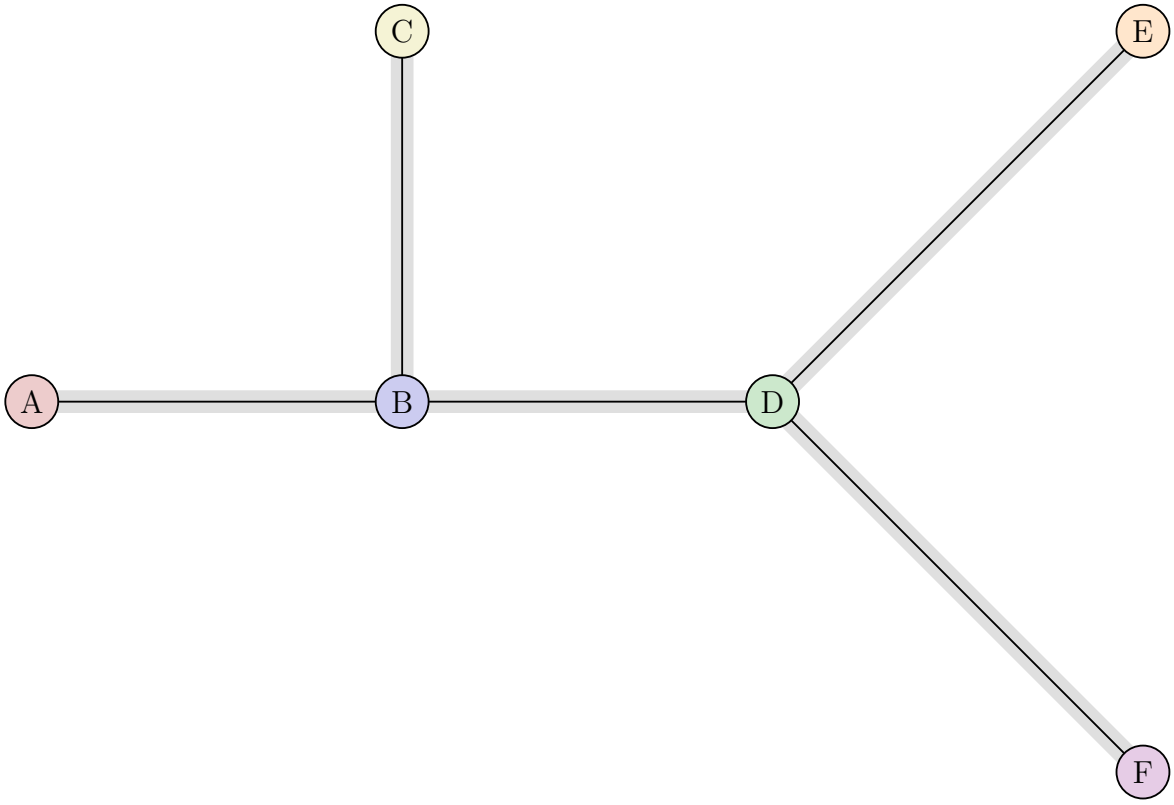
$A \to B \to A \to B \to D \to F \to D \to E \longrightarrow$

$A \to B \to D \to F \to D \to E \longrightarrow$

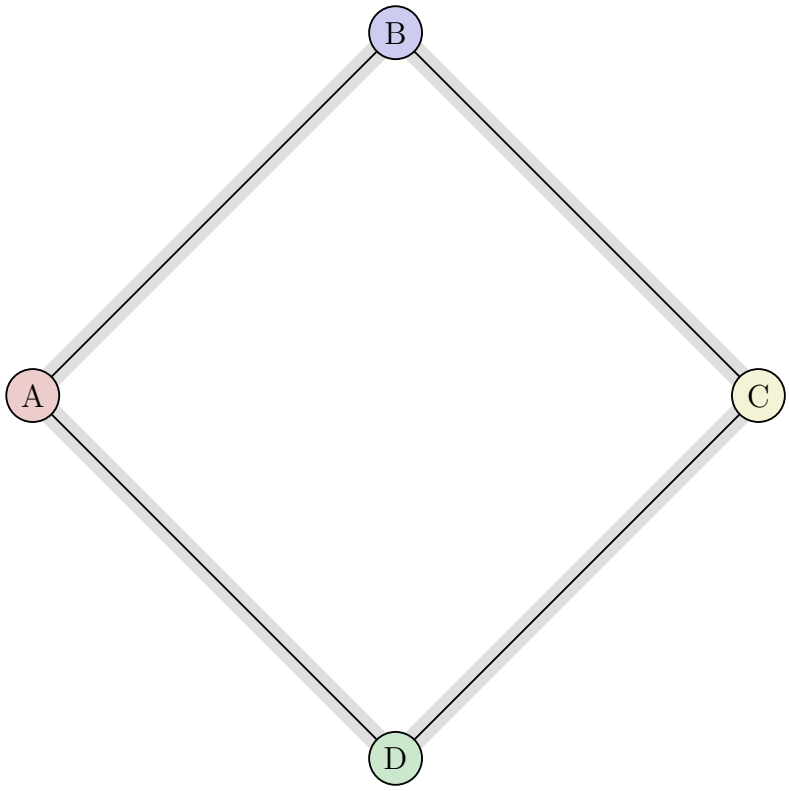
$C \to B \to C \to B \to D \to F \longrightarrow$



Problem 2: Make three different 12-step trips from A back to A, each using every road. Can any trip home on this map survive after every immediate reversal is removed?



Problem 3: Make every different trip from A back to A that can remain after shortening an eight-step ring trip. Find the fewest road steps in a surviving trip home.



Problem 4: Remove different first detours from the recorded trip below. Can two removal orders end differently?

$$A \to B \to A \to B \to C \to B \to A \to D \to C \to D \to A \quad \longrightarrow _____$$